

Time-series Foundation Model Evaluation

Executive Summary

Thesis project

Evidence checked through 29 August 2026

Decision snapshot

Chronos-2 beats the best simple baseline on 153 of 189 completed full-profile tasks, but persistence remains best on all three Exchange Rate settings. The control effects are not universal, identity covariates are an oracle capability check, and the complete foundation grid splits task wins and mean rank between TS-ICL and Chronos-2.

1 Evidence inventory

Full-profile family	Selected inputs	Tasks
Univariate	945	189
Controls	1,512	189
Covariates	756	189
Foundation models	512	128

The first three synchronized reports use finite CSV panels. Consequently the new default NaN-to-zero loading policy is inert for those completed computations. The foundation replacement records 21 zero-filled missing values in each Weather configuration and completes the remaining eligible data.

2 Univariate evidence

Chronos-2 has lower aggregate nMSE than the best repeat, lookback, or mean baseline on 153 of 189 tasks, with one tie. Its median relative improvement is 24.10%. Across the nine Electricity, Solar, and Traffic publication settings, the improvement ranges from 20.88% to 51.26%. Exchange Rate is the important counterexample: persistence is better at all three settings, by 0.60% to 4.88%. Raw averages across all datasets are not reported because several TIME tasks have near-zero normalization denominators and would dominate them.

3 Controls and covariates

Removing constant windows improves only 5 of 189 matched Chronos-2 tasks and has zero median effect, although those five avoid extreme normalized-error cases. Instance normalization improves 93 of 189 tasks and also has zero median effect. Neither intervention is supported as a universal default.

The identity-covariate capability control beats the same model without covariates on 185 of 189 Chronos-2 tasks and 176 of 189 TS-ICL tasks. Median relative nMSE improvements are 97.93% and 98.07%, respectively. Identity passes the target itself as a covariate, so this establishes adapter capability rather than a deployable forecasting gain.

The synchronized covariate raw results are eligible, but the old report stage grouped by covariate mode and then searched each group for a simple baseline. Because that family contains only Chronos-2 and TS-ICL, the comparison tables were empty. The corrected report pairs each augmented row with the same backbone and task under `covariate_mode=none`; no inference rerun is required.

4 Foundation-model comparison

Replacement job 2964473 completed 512 rows for 128 tasks, four models, and 64 datasets, superseding the failed evaluation and partial report. TS-ICL has the lowest nMSE on 51 tasks, Chronos-2 on 50, Chronos-Bolt on 17, and Chronos-T5 on 10. Chronos-2 nevertheless has the best mean rank and beats TS-ICL head-to-head on 69 of 128 tasks. The grid therefore identifies two strong leaders under different summaries rather than one universal winner.

5 Next evidence

Deploy the report fix and rerun only the full covariate report stage. The foundation evaluation and report are complete and require no rerun.